

THAKUR VAIBHAV SINGH

Biomedical Engineer | AIR 154 GATE BM 2026

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Portfolio: <https://vaibhavthkr.github.io/portfolio/>

PROFESSIONAL SUMMARY

Final-year Biomedical Engineering student (Osmania University) with hands-on experience in medical device design, Computer Aided Design (Solidworks, Abaqus), Biomechanics, embedded systems, Biomedical signal processing, and wearable sensor integration. GATE BM 2026 AIR 154 demonstrates strong analytical foundation. Proven ability to work in cross-functional teams (IEEE, TEDx, national symposia), deliver technical documentation, and apply engineering principles to real-world healthcare challenges.

EXPERIENCE

Intern - Biomedical Sensors and Systems Lab, University of Memphis, USA

May 2026 - Present

- Working on Research projects involving Biomechanics of Elbow joint, Knee joint, Shoulder joint, using softwares like OpenSim, Mimics, Python programming

PROJECTS

A Deep Learning based Real-Time Estimation of Tibiofemoral Contact Forces using Wearable IMU Sensors and Personalized Neuromusculoskeletal Modelling

- Currently working on it as Academic mini project
- Estimate real time knee joint contact forces based upon the knee joint angle using IMU sensors and NMSM modelling.

Fabrication of PCL nanofiber using air blown method

- Fabricated PCL Nanofibers
- Used Air compressor to generate high pressure to spray PCL Solution

Low cost portable tribometer

- Made a simple pin on disk tribometer.
- Integrated speed control and timer for the stepper motor, the deflections from the stylus are recorded through the load cell and drafted technical specifications and test plans to record data in Excel.

Computer vision based upper limb rehabilitation system

- Made a computer vision-based GUI using python for rehabilitation of wrist joint, elbow joint and shoulder joint.
- It consists of demo video of exercise and real time visual and audio feedback, reps counter, exercise history.

Real-time gait parameters monitoring via MPU6050 and ESP32

- Monitored various parameters such as Heel strike, stance period, toe off etc.
- The system was wireless just attach on the Heel and use, data was plotted in MATLAB

Gamified computer vision-based palm rehabilitation system

- The patients can play Dino game using their palm
- This project helps gamification of Rehabilitation systems; further research could involve designing personalized gamified Rehabilitation systems

EDUCATION

University College of Engineering (A), Osmania University

Hyderabad, TG

B.E Biomedical Engineering

August 2023 – June 2027

Relevant Coursework: Biomedical Equipment; Biomedical Signal Processing, Biomedical Image processing, Biomedical Imaging systems, Biomaterials, Biomedical Equipment; Medical Device Regulations

Sri Chaitanya Junior College

ECIL, Hyderabad

Mathematics, Physics, Chemistry

April 2020 – May 2023

Nalanda High School

Neredmet, Hyderabad

SSC

April 2020

PUBLICATIONS

Published: Paper presented at IFIP IoT'25:

"A Multimodal Control Approach for Smart Wheelchairs Using Accelerometer and EMG Gesture Interfaces" — demonstrates R&D contribution and technical communication.

SKILLS

Engineering Software & Simulation: OpenSim, MATLAB, Python, Embedded C, SolidWorks, Mimics, Abaqus, Proteus, Arduino IDE

Electronics & Embedded Systems: ESP32, Arduino Uno, IoT (Internet of Things), Biomedical Sensors, Embedded system design, Wearable devices.

Core Competencies: Biomechanics, Biomedical Signal Processing, Biomedical Equipment Management, Medical Image processing, Medical Imaging systems, Biotribology, Biomaterials, 3D printing, Medical Image segmentation and 3D modelling, Medical Device regulations

Soft Skills: Technical Documentation, Cross-functional Team Collaboration, Problem Solving, Analytical Thinking, Presentation, Project Coordination, Verification & Validation (V&V), Root Cause Analysis, and GD&T.

ACTIVITIES

IEEE SB OUCE EMBS

Organizer

Apr 2025 – Apr 2026

- Coordinated logistics and implemented lean process workflows to optimize event management for 10+ events.

IEEE SB OUCE

Design Lead

Apr 2025 - Apr 2026

- Managing and guiding design team of 12 design volunteers for graphic design.

National Symposium on Instrumentation - 45, ISOI Bangalore

Volunteer

Dec 2025 - Jan 2026

- Lead the Design team, delivered Brochure, invitations, various posters, certificates, ID Cards.
- Stage management and ensured smooth flow of the event

TEDxOsmaniaU 2025

Committee Member

Aug 2025 – Nov 2025

- Received the Letter of Recommendation from the Managing Director, TEDxOsmaniaU 2025
- Handled Stage management during the main event and also the Theme reveal event of the TEDxOsmaniaU 2025.

Section Student Congress 2025, IEEE Hyderabad Section

Volunteer

Oct 2024

- Volunteered in logistics of the event

ACHIEVEMENTS/ CO CURRICULAR ACTIVITIES

- Achieved GATE 2026 **AIR 154** in Biomedical Engineering paper. [Score card](#)
- Volunteered in organizing National Level Hands on Workshop on Building Smart Biomedical Devices using IOT on 20th – 21st February 2026.
- Won 1st prize in Poster presentation competition with my teammate Ushasree B, presented “Mitigating SANS A Novel Wearable Device for Real-Time Non-Invasive ICP Monitoring in Spaceflight” in National Symposium on Instrumentation-45 organized by Department of Biomedical Engineering, UCE (A), OU in collaboration with ISOI, IISc Bangalore. [Certificate link](#)
- Industry visit to AMTZ, Vishakhapatnam.
- Attended Section Student Congress organized by IEEE Hyderabad section. [Certificate link](#)
- Attended Hands on training on Biomedical Equipment organized by department of Biomedical Engineering, UCE (A), OU.
- Won 1st Prize in Poster presentation competition, presented “Thermal vision-based Sleep apnea detection using CNN and LSTM networks” in Nation level Technical Symposium organized by Spoorthy Engineering college, Hyderabad. [Certificate link](#)
- Participated in the National Seminar on Recent Trends in Medical Equipment – 24 (RTME-24) conducted by Department of Biomedical Engineering, University College of Engineering (A), Osmania University. [Certificate link](#)
- Won 1st prize in Quiz competition in Hands on Workshop on Biomaterials and Fabrication Techniques organized by Department of Biomedical Engineering, UCE (A), OU. [Certificate link](#)